

Resolution in Propositional Logic and First-Order Logic

Resolution in Artificial Intelligence

- Resolution is a rule of inference used to automatically prove logical statements.
- Resolution reduces logical reasoning to a mechanical procedure. That is why it is powerful for AI. Computers can apply it repeatedly without human creativity.
- This method is basically used for proving the satisfiability of a sentence.
- In resolution method, we use “Proof by Refutation” technique to prove the given statement.

It forms the foundation of:

- Automated theorem proving
- Logic programming (e.g., Prolog)
- Knowledge-based systems

Resolution in Artificial Intelligence

- The Key idea for the resolution method is to use the knowledge base and negated goal to obtain null clause (which indicates contradiction).
- Resolution method is also called **Proof by Refutation**. Since the knowledge base itself is consistent, the contradiction must be introduced by a negated goal.
- As a result , we have to conclude that the original goal is true.

Resolution in Artificial Intelligence

The Main Objective

Given:

$$KB \models \alpha$$

Means: “KB entails alpha.”

OR

“Alpha logically follows from the knowledge base.”

OR

“KB logically entails alpha.”

Entailment → Unsatisfiability checking

Resolution in Artificial Intelligence

The Main Objective

Given:

$$KB \models \alpha$$

What Does It Mean?

It means:

In every interpretation (model) where KB is true, α is also true.

So: If KB is true $\rightarrow \alpha$ must also be true.

There is no situation where KB is true and α is false.

- We want to prove that α logically follows from KB.
- Instead of proving directly, we use **Proof by Refutation**.

Proof by Refutation

We prove:

$$KB \wedge \neg\alpha$$

is **unsatisfiable**.

If contradiction ($\perp$) is derived:

$\rightarrow \alpha$ must be true.

Why?

Because KB is assumed consistent.

➤ If adding $\neg\alpha$ causes contradiction, the contradiction must be due to $\neg\alpha$. Hence α must follow.

Resolution in Propositional Logic (PL)

- In propositional logic, the resolution method is only the reference rule which gives a new clause when two or more clauses are coupled together.
- Resolution is the only inference rule needed for PL theorem proving.

Core Rule:

$$(A \vee B) \wedge (\neg B \vee C) \Rightarrow (A \vee C)$$

Important Terminology

Literal = Atomic proposition or its negation

Clause = Disjunction of literals

CNF = Conjunction of clauses a.k.a **Conjunctive Normal Form (AND of ORs)**

Example:

$$(\neg P \vee Q) \wedge (P \vee R)$$

- Resolution operates only on clauses. That is why CNF conversion is mandatory.
- Similarly, DNF = **Disjunctive Normal Form (OR of ANDs)**

Steps to Convert PL to CNF

1. Remove implications

$$P \rightarrow Q \equiv \neg P \vee Q$$

2. Move negations inward

(Using De Morgan's laws)

3. Distribute OR over AND

4. Simplify

Steps to Convert PL to CNF

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4. Simplify

Example:

$P \vee (Q \wedge R)$

becomes:

$(P \vee Q) \wedge (P \vee R)$

Resolution Algorithm (PL)

1. Convert KB to CNF
 2. Add negation of goal
 3. Select two clauses with complementary literals
 4. Generate resolvent
 5. Add resolvent to clause set
 6. Repeat
 7. If empty clause found $\rightarrow$ Success
- Empty clause ($\square$ or $\perp$) means contradiction.
 - If no new clause can be generated $\rightarrow$ proof fails.

Resolution Algorithm (PL)

What is a Resolvent?

A resolvent is the new clause obtained after applying the resolution rule to two clauses containing complementary literals.

If we have:

$$(L \vee X) \wedge (\neg L \vee Y) \Rightarrow (X \vee Y)$$

Then:

- L and $\neg L$ are complementary literals.
- After removing them,
- The remaining literals combine to form the **resolvent**.

So:

Resolvent = result of resolution.

PL Example

Given:

1. $P \rightarrow Q$

2. $Q \rightarrow R$

3. P

Prove: R

Step 1: Convert to CNF

Remove implications:

$$P \rightarrow Q \equiv \neg P \vee Q$$

$$Q \rightarrow R \equiv \neg Q \vee R$$

So we get clauses:

1. $\neg P \vee Q$

2. $\neg Q \vee R$

3. P

Add negation of goal:

4. $\neg R$

Now our clause set is:

- $\neg P \vee Q$

- $\neg Q \vee R$

- P

- $\neg R$

PL Example

Given:

1. $P \rightarrow Q$

2. $Q \rightarrow R$

3. P

Prove: R

Step 2: First Resolution

Take:

Clause 1: $\neg P \vee Q$

Clause 3: P

Complementary literals:

P and $\neg P$

Remove them.

Remaining literal:

Q

So:

Resolvent 1 = Q

PL Example

Given:

1. $P \rightarrow Q$

2. $Q \rightarrow R$

3. P

Prove: R

Step 3: Second Resolution

Now use:

Resolvent: Q

Clause 2: $\neg Q \vee R$

Complementary literals:

Q and $\neg Q$

Remove them.

Remaining literal:

R

So:

Resolvent 2 = R

PL Example

Given:

1. $P \rightarrow Q$

2. $Q \rightarrow R$

3. P

Prove: R

Step 4: Final Resolution

Now resolve:

R

$\neg R$

Complementary literals:

R and $\neg R$

Remove them.

Nothing remains.

This gives:

$\perp$ (Empty clause)

This empty clause is also a **resolvent**.

PL Example

Why Is Empty Clause Important?

Given:

1. $P \rightarrow Q$

2. $Q \rightarrow R$

3. P

Prove: R

Empty clause means:

False.

We derived contradiction.

Since contradiction arose after adding $\neg R$:

→ Conclusion: R must be logically entailed.

Track all resolvents:

1. From $(\neg P \vee Q)$ and $P \rightarrow Q$

2. From $(\neg Q \vee R)$ and $Q \rightarrow R$

3. From R and $\neg R \rightarrow \perp$

Each new derived clause is called a **resolvent**.

Why Resolution is Complete (PL)

Theorem:

If a set of propositional clauses is unsatisfiable, resolution will derive the empty clause.

Therefore:

Resolution is:

- Sound
- Complete
- Decidable

Resolution in First-Order Logic (FOL)

PL limitations:

- No variables
- No quantifiers
- Cannot represent general rules

Example:

$$\forall x (Human(x) \rightarrow Mortal(x))$$

Requires FOL.

Why PL Resolution Fails in FOL

In PL:

Human(Socrates)

In FOL:

Human(x)

These are not syntactically identical.

We need substitution.

This leads to:

Unification

What is Unification?

Unification is the process of finding a substitution that makes two logical expressions identical.

In simple words:

It tells us **how to replace variables** so that two predicates match exactly.

“Unification is like solving equations between predicates. It figures out what values variables must take so that two statements become identical.”

Why Do We Need Unification?

In Propositional Logic:

- P and $\neg P$ match directly.

In First Order Logic:

- $\text{Human}(x)$
- $\text{Human}(\text{Socrates})$

These are not syntactically identical.

We must replace:

$x \rightarrow \text{Socrates}$

to make them match.

That replacement process is called **Unification**.

Why Do We Need Unification?

Formal Definition

Two literals are unifiable if there exists a substitution θ such that:

$$L_1\theta = L_2\theta$$

θ is called a **unifier**.

The best one is called:

Most General Unifier (MGU).

Why Do We Need Unification?

Example 1

Unify:

Knows(John, x)

Knows(John, Mary)

Substitution:

$\theta = \{ x \rightarrow \text{Mary} \}$

After substitution:

Knows(John, Mary)

Knows(John, Mary)

They match.

Unification Example

Example 2

Unify:

$P(x, y)$

$P(a, b)$

Substitution:

$\theta = \{ x \rightarrow a, y \rightarrow b \}$

Example 3

Unify:

$P(x, f(y))$

$P(a, f(b))$

Compare arguments one by one:

x must be a

y must be b

So:

$\theta = \{ x \rightarrow a, y \rightarrow b \}$

Unification Example

When Unification Fails

Unify:

$P(x, x)$

$P(a, b)$

This would require:

$x = a$

$x = b$

Which means:

$a = b$

If $a \neq b \rightarrow$ unification fails.

Why “Most General” Unifier?

Consider:

$P(x)$

$P(a)$

Possible substitutions:

$x \rightarrow a$

$x \rightarrow a$ and add extra unnecessary substitutions

The simplest substitution is:

$\theta = \{ x \rightarrow a \}$

This is called the **Most General Unifier (MGU)**.

It makes minimal commitments.

How Unification Works in Resolution

Resolution rule in FOL:

$$(P(x) \vee R) \wedge (\neg P(y) \vee S) \Rightarrow (R \vee S)\theta$$

Before resolving:

1. Find substitution θ that makes $P(x)$ and $P(y)$ identical.
2. Apply θ .
3. Then eliminate complementary literals.
4. Combine remaining literals.

How Unification Works in Resolution

Example:

Given:

$\text{Human}(x) \rightarrow \text{Mortal}(x)$
 $\text{Human}(\text{Socrates})$

Convert:

$\neg \text{Human}(x) \vee \text{Mortal}(x)$
 $\text{Human}(\text{Socrates})$

To resolve:

$\text{Human}(\text{Socrates})$
 $\neg \text{Human}(x)$

Unify:

$\theta = \{ x \rightarrow \text{Socrates} \}$

Now they match.

After removing complementary pair:

$\text{Mortal}(\text{Socrates})$

Full Pipeline for FOL Resolution

1. Remove implications
2. Move negations inward
3. Standardize variables
4. Skolemization
5. Drop universal quantifiers
6. Convert to CNF
7. Apply resolution with unification

Skolemization

Purpose: Remove existential quantifiers.

Example:

$$\exists x P(x)$$

Replace x with constant a :

$$P(a)$$

If dependent:

$$\forall y \exists x P(x, y)$$

Replace with:

$$P(f(y), y)$$

- Skolem functions preserve satisfiability, not equivalence.
- This is enough for refutation proof.

FOL Resolution Rule

$$(P(x) \vee R) \wedge (\neg P(y) \vee S) \Rightarrow (R \vee S)\theta$$

θ = Most General Unifier (MGU)

Unification

Unification = Finding substitution θ such that:

Two literals become identical.

Example:

Knows(John, x)

Knows(John, Mary)

$\theta = \{ x \rightarrow \text{Mary} \}$

Complete FOL Example

Given:

$\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$

$\text{Human}(\text{Socrates})$

Prove:

$\text{Mortal}(\text{Socrates})$

Convert:

$\neg \text{Human}(x) \vee \text{Mortal}(x)$

$\text{Human}(\text{Socrates})$

Add:

$\neg \text{Mortal}(\text{Socrates})$

Resolution with Unification

Unify:

Human(Socrates)

$\neg$ Human(x)

$\theta = \{ x \rightarrow \text{Socrates} \}$

Generate:

Mortal(Socrates)

Resolve with:

$\neg$ Mortal(Socrates)

$\rightarrow \perp$

Proof complete.

Theoretical Properties (FOL)

- Sound
- Refutation-complete
- Semi-decidable

If statement true $\rightarrow$ proof found

If false $\rightarrow$ may loop forever

Reason:

FOL validity is undecidable.

PL vs FOL Comparison

Feature	PL	FOL
Variables	No	Yes
Unification	Not required	Required
Skolemization	Not required	Required
Decidable	Yes	No
Expressiveness	Limited	Powerful

Practice Problems

1. $(P \rightarrow Q), (Q \rightarrow R), P \models R$

2. $\forall x (Cat(x) \rightarrow Animal(x)), Cat(Tom) \models Animal(Tom)$

3. $\forall x (Man(x) \rightarrow Mortal(x)), \exists x Man(x) \models \exists x Mortal(x)$

Example OF Propositional Resolution

- Consider the following Knowledge Base:
 1. The humidity is high or the sky is cloudy.
 2. If the sky is cloudy, then it will rain.
 3. If the humidity is high, then it is hot.
 4. It is not hot.
- **Goal:** It will rain.
- Use propositional logic and apply resolution method to prove that the goal is derivable from the given knowledge base.



Solution:

- Let's construct propositions of the given sentences one by one:
- The humidity is high or the sky is cloudy. 1
Let, P: Humidity is high.
Q: Sky is cloudy.
It will be represented as **$P \vee Q$** .
- If the sky is cloudy, then it will rain. 2
Q: Sky is cloudy. ...**from(1)**
Let, R: It will rain.
It will be represented as **$Q \rightarrow R$** .

- 
- If the humidity is high, then it is hot.

.... 3

P: Humidity is high. ...**from(1)**

Let, S: It is hot.

It will be represented as **P** $\rightarrow$ **S**.

- It is not hot.

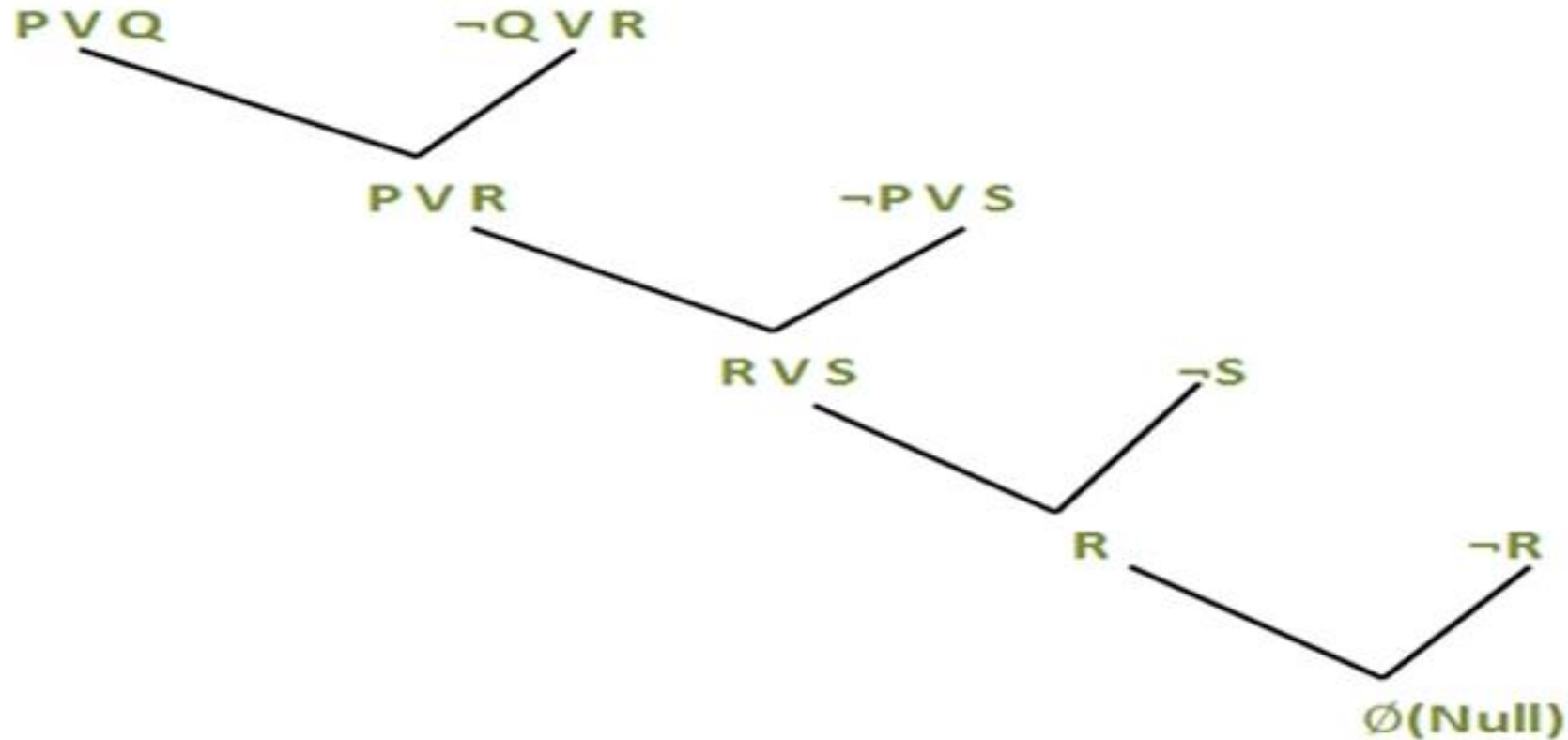
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$\neg$ S: It is not hot.

Applying resolution method

- Convert the given axiom into clausal form, i.e., disjunction form.
- Apply and prove the given goal using negation rule.
- Use those literals which are needed to prove.
- Solve the clauses together and achieve the goal.
- In (2), $Q \rightarrow R$ will be converted as $(\neg Q \vee R)$
- In (3), $P \rightarrow S$ will be converted as $(\neg P \vee S)$
- **Negation of Goal ($\neg R$):** It will not rain.

Finally, apply the rule as shown below:



After applying Proof by Refutation (Contradiction) on the goal, the problem is solved, and it has terminated with a **Null clause ($\emptyset$)**. Hence, the goal is achieved. Thus, It is not raining.

Example:

Proof with Resolution

Given the following hypotheses:

1. If it rains, Joe brings his umbrella ($r \rightarrow u$)
2. If Joe has an umbrella, he doesn't get wet ($u \rightarrow \text{NOT } w$)
3. If it doesn't rain, Joe doesn't get wet ($\text{NOT } r \rightarrow \text{NOT } w$)

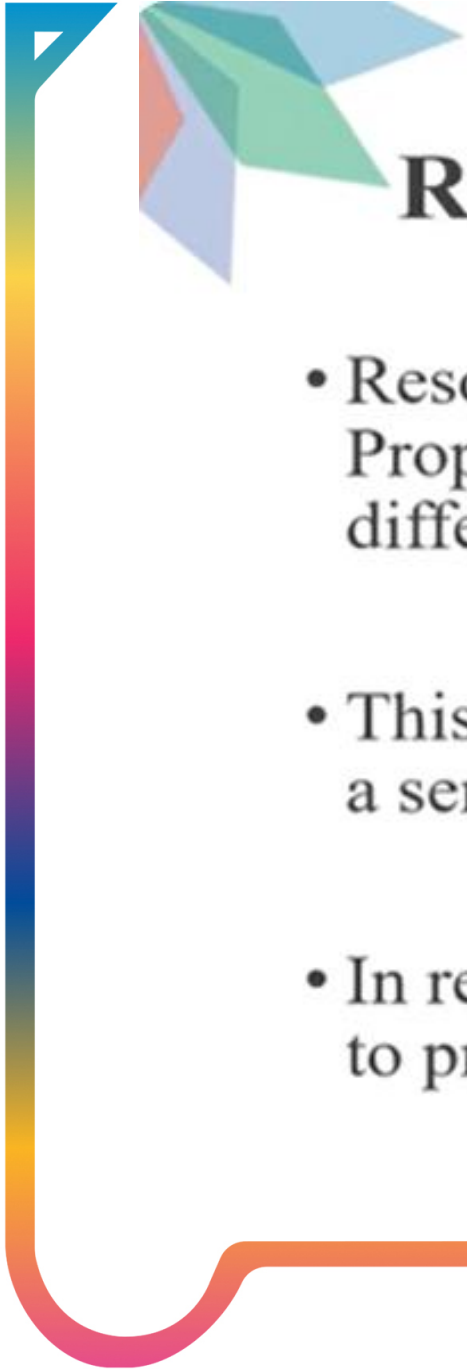
prove that Joe doesn't get wet ($\text{NOT } w$)

We first put each hypothesis in CNF:

1. $r \rightarrow u \equiv (\text{NOT } r \text{ OR } u)$
2. $u \rightarrow \text{NOT } w \equiv (\text{NOT } u \text{ OR } \text{NOT } w)$
3. $\text{NOT } r \rightarrow \text{NOT } w \equiv (r \text{ OR } \text{NOT } w)$

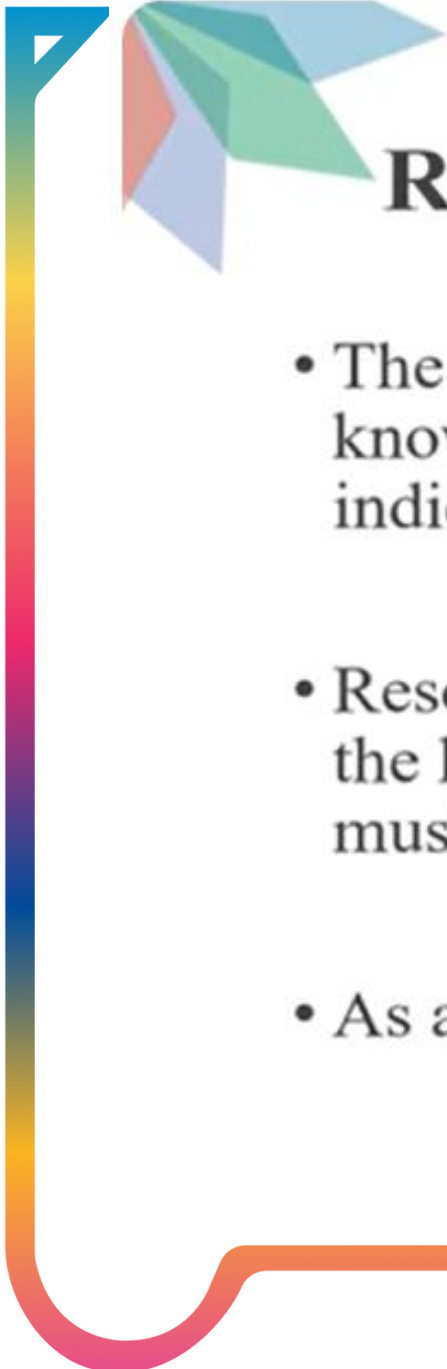
We then use resolution on the hypotheses to derive the conclusion (NOT w):

- | | |
|-------------------|--------------------|
| 1. NOT r OR u | Premise |
| 2. NOT u OR NOT w | Premise |
| 3. r OR NOT w | Premise |
| 4. NOT r OR NOT w | L1, L2, resolution |
| 5. NOT w OR NOT w | L3, L4, resolution |
| 6. NOT w | L5, idempotence |
| 7. QED | |

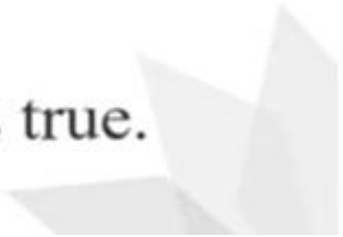


Resolution Method in AI

- Resolution method is an inference rule which is used in both Propositional as well as First-order Predicate Logic in different ways.
- This method is basically used for proving the satisfiability of a sentence.
- In resolution method, we use **Proof by Refutation** technique to prove the given statement.



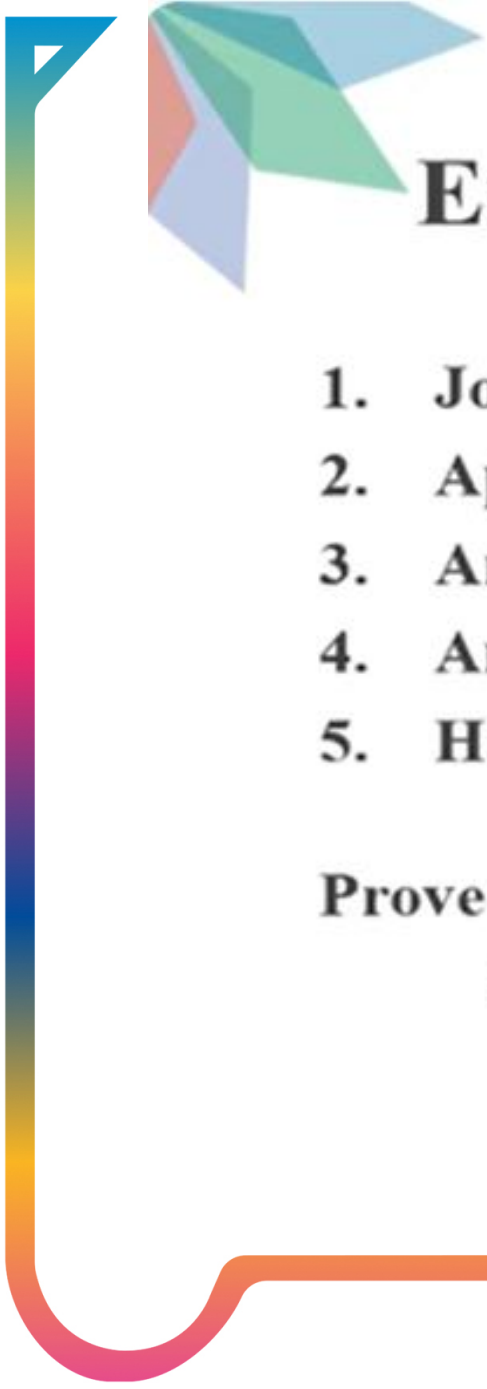
Resolution Method in AI

- The key idea for the resolution method is to use the knowledge base and negated goal to obtain null clause(which indicates contradiction).
 - Resolution method is also called **Proof by Refutation**. Since the knowledge base itself is consistent, the contradiction must be introduced by a negated goal.
 - As a result, we have to conclude that the original goal is true.
- 



Resolution Method in FOPL/Predicate Logic

- Resolution method in FOPL is an uplifted version of propositional resolution method.
- **In FOPL, the process to apply the resolution method is as follows:**
 - Conversion of facts into first-order logic.
 - Convert FOL statements into CNF
 - Negate the statement which needs to prove (proof by contradiction)
 - Draw resolution graph (unification).



Example

1. John likes all kind of food.
2. Apple and vegetable are food
3. Anything anyone eats and not killed is food.
4. Anil eats peanuts and still alive
5. Harry eats everything that Anil eats.

Prove by resolution that:
John likes peanuts.

Step-1: Conversion of Facts into FOL

- a) John likes all kind of food.
- b) Apple and vegetable are food
- c) Anything anyone eats and not killed is food.
- d) Anil eats peanuts and still alive
- e) Harry eats everything that Anil eats.

Prove by resolution that:
John likes peanuts.

- a. $\forall x: \text{food}(x) \rightarrow \text{likes}(\text{John}, x)$
 - b. $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
 - c. $\forall x \forall y: \text{eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \text{food}(y)$
 - d. $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$.
 - e. $\forall x : \text{eats}(\text{Anil}, x) \rightarrow \text{eats}(\text{Harry}, x)$
 - f. $\forall x: \neg \text{killed}(x) \rightarrow \text{alive}(x)$
 - g. $\forall x: \text{alive}(x) \rightarrow \neg \text{killed}(x)$
 - h. $\text{likes}(\text{John}, \text{Peanuts})$
- } added predicates.



Step-2: Conversion of FOL into CNF

1. Eliminate all implication ($\rightarrow$) and rewrite
2. Move negation ($\neg$) inwards and rewrite
3. Rename variables or standardize variables to unique variable.
4. Move all **quantifiers** to the left without changing their relative order.
5. Eliminate existential instantiation quantifier by elimination $\exists$ (Skolemization).
6. Drop Universal quantifiers $\forall$.
7. Convert the formula into a **conjunction of disjuncts**.

Step-2: Conversion of FOL into CNF

Eliminate all implication ($\rightarrow$) and rewrite : $A \rightarrow B$ with $\neg A \vee B$.

- a. $\forall x: \text{food}(x) \rightarrow \text{likes}(\text{John}, x)$
 - b. $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
 - c. $\forall x \forall y: \text{eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \text{food}(y)$
 - d. $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
 - e. $\forall x: \text{eats}(\text{Anil}, x) \rightarrow \text{eats}(\text{Harry}, x)$
 - f. $\forall x: \neg \text{killed}(x) \rightarrow \text{alive}(x)$
 - g. $\forall x: \text{alive}(x) \rightarrow \neg \text{killed}(x)$
 - h. $\text{likes}(\text{John}, \text{Peanuts})$
- } **added predicates.**
- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
 - $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
 - $\forall x \forall y \neg [\text{eats}(x, y) \wedge \neg \text{killed}(x)] \vee \text{food}(y)$
 - $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
 - $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
 - $\forall x \neg [\neg \text{killed}(x)] \vee \text{alive}(x)$
 - $\forall x \neg \text{alive}(x) \vee \neg \text{killed}(x)$
 - $\text{likes}(\text{John}, \text{Peanuts})$

Step-2: Conversion of FOL into CNF

Move negation ($\neg$) inwards and rewrite

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
 - $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
 - $\forall x \forall y \neg [\text{eats}(x, y) \wedge \neg \text{killed}(x)] \vee \text{food}(y)$
 - $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
 - $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
 - $\forall x \neg [\neg \text{killed}(x)] \vee \text{alive}(x)$
 - $\forall x \neg \text{alive}(x) \vee \neg \text{killed}(x)$
 - $\text{likes}(\text{John}, \text{Peanuts})$
- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
 - $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
 - $\forall x \forall y \neg \text{eats}(x, y) \vee \text{killed}(x) \vee \text{food}(y)$
 - $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
 - $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
 - $\forall x \text{killed}(x) \vee \text{alive}(x)$
 - $\forall x \neg \text{alive}(x) \vee \neg \text{killed}(x)$
 - $\text{likes}(\text{John}, \text{Peanuts})$

Step-2: Conversion of FOL into CNF

Rename variables or standardize variables

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall x \forall y \neg \text{eats}(x, y) \vee \text{killed}(x) \vee \text{food}(y)$
- $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
- $\forall x \neg \text{killed}(x) \vee \text{alive}(x)$
- $\forall x \neg \text{alive}(x) \vee \neg \text{killed}(x)$
- $\text{likes}(\text{John}, \text{Peanuts})$
- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall y \forall z \neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
- $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall w \neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$
- $\forall g \neg \text{killed}(g) \vee \text{alive}(g)$
- $\forall k \neg \text{alive}(k) \vee \neg \text{killed}(k)$
- $\text{likes}(\text{John}, \text{Peanuts})$



Step-2: Conversion of FOL into CNF

Eliminate existential instantiation quantifier by elimination.

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall y \forall z \neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
- $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall w \neg \text{eats}(\text{Anil}, \mathbf{w}) \vee \text{eats}(\text{Harry}, \mathbf{w})$
- $\forall g \neg \text{killed}(g) \vee \text{alive}(g)$
- $\forall k \neg \text{alive}(k) \vee \neg \text{killed}(k)$
- $\text{likes}(\text{John}, \text{Peanuts}).$

Step-2: Conversion of FOL into CNF

Drop Universal quantifiers.

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
 - $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
 - $\forall y \forall z \neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
 - $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
 - $\forall w \neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$
 - $\forall g \neg \text{killed}(g) \vee \text{alive}(g)$
 - $\forall k \neg \text{alive}(k) \vee \neg \text{killed}(k)$
 - $\text{likes}(\text{John}, \text{Peanuts})$.
- $\neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
 - $\text{food}(\text{Apple})$
 - $\text{food}(\text{vegetables})$
 - $\neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
 - $\text{eats}(\text{Anil}, \text{Peanuts})$
 - $\text{alive}(\text{Anil})$
 - $\neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$
 - $\text{killed}(g) \vee \text{alive}(g)$
 - $\neg \text{alive}(k) \vee \neg \text{killed}(k)$
 - $\text{likes}(\text{John}, \text{Peanuts})$.

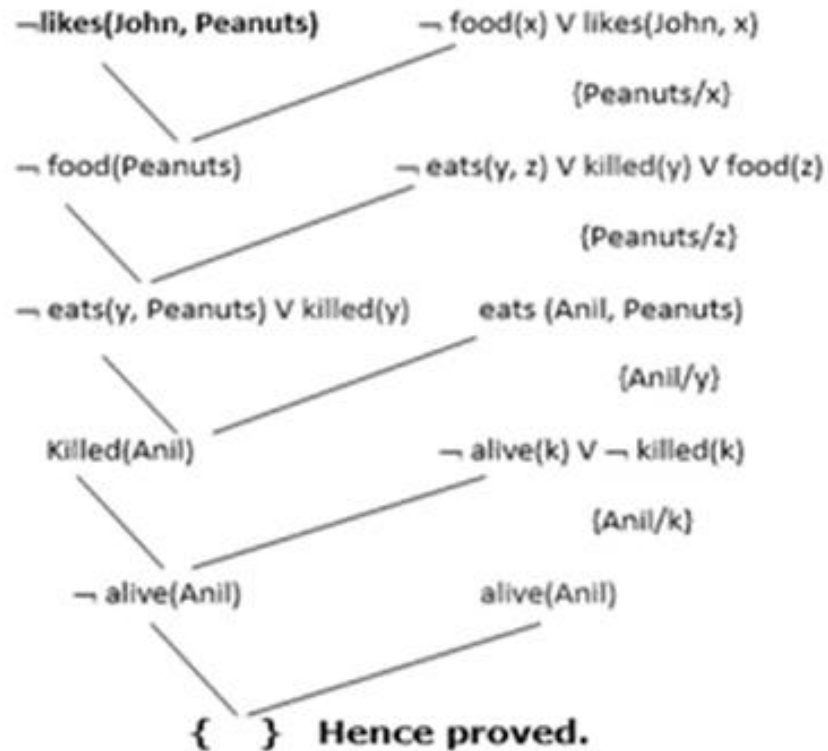


Step-3: Negate the statement to be proved

- In this statement, we will apply negation to the conclusion statements, which will be written as

$\neg \text{likes}(\text{John}, \text{Peanuts})$

Draw Resolution graph



- $\neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple})$
- $\text{food}(\text{vegetables})$
- $\neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
- $\text{eats}(\text{Anil}, \text{Peanuts})$
- $\text{alive}(\text{Anil})$
- $\neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$
- $\text{killed}(g) \vee \text{alive}(g)$
- $\neg \text{alive}(k) \vee \neg \text{killed}(k)$
- $\text{likes}(\text{John}, \text{Peanuts})$.

Practice Question

Prove the following statements using resolution in predicate logic:

Statements:

1. All humans are mortal.
2. Socrates is a human.

Conclusion: Therefore, Socrates is mortal.

Use predicate logic and the resolution principle to derive the conclusion.



<https://www.javatpoint.com/ai-resolution-in-first-order-logic>

<https://www.youtube.com/watch?v=Cbl-Q2a5rUU>